Module Description

IN8027: Introduction to Informatics for Students of Management & Technology – Programming Lab Course

Department of Computer Science

Module level:	Language:	Module duration:	Occurrence:
Bachelor	English	one semester	winter semester
Credits*:	**Total number of hours:**	**Self-study hours:**	**Contact hours:**
5	150	120	30

* The number of credits can vary depending on the corresponding SPO version. The valid number is always indicated on the Transcript of Records or the Performance Record.

Description of achievement and assessment methods:
Type of assessment: individual grading of submitted written exercise work ("Übungsleistung / sonstige schriftliche Leistung" according to TUM-APSO): short project reports and project artefacts (code, software engineering process documents (e.g. UML diagrams) etc.).
During the four-week block period, students submit their intermediate and final work results (especially including their written program-code and documents of the software engineering process) electronically via a revision control system (usually GIT). Furthermore, at the end of the block period, each student submits a small, concise project report, in which her individual contributions to the work results are described.
The submitted written exercise work documents the student's degree of acquaintance with the programming language Java and their practical skills in terms of programming in the small and allows to assess how well the students are able to apply database-systems and SQL, basic object-oriented programming and Java, and basic algorithms and data structures for solving small to medium sized programming problems. The submitted exercise work also shows how well the participants are acquainted with and can apply the basics of a modern agile software development process.
Individual project reports, the documented submission history in the revision control environment, and the documents created in the structured software development process ensure that student contributions may be assessed on an individual basis.
The retake exam is offered in the form of a written exam (120 minutes, closed book) at the end of the semester.

Possibility of re-taking:
In the next semester: No
At the end of the semester: Yes

(Recommended) requirements:
None.
Participants should attend the module IN8005: "Introduction to Informatics for Students of Management and Technology" in the same semester.

Contents:
• Object-oriented software development with Java
• SQL integration in Java
• Agile software development processes (typically Scrum)
• Revision control systems (typically GIT)

Study goals:
Upon successful completion of the module, participants are acquainted with the programming language Java and master programming in the small. Participants are able to apply the contents taught in the module IN8005 (foundations of database-systems and SQL, foundations of object-oriented programming and Java, foundations of algorithms and data structures) for solving small to medium sized programming problems in their professional field and/or for later scientific work. Participants are acquainted with and can apply the basics of a modern agile software development process (typically Scrum) for the development of solutions to these problems and are able to collaborate with informatics professionals in analyzing and evaluating the complexity of possible software solutions for professional problem settings.
Students are able to complete the tasks of their project in a team environment. They solve the given task by constructively and conceptually collaborating in a team. They are able to integrate involved persons into the various tasks considering the group situation. Furthermore, the students are able to conduct solution processes through constructively and conceptually acting in a team.

Teaching and learning methods:
The lab-course takes the form of a four-week block lab-course taking place in the second half of the semester. In the first half of the semester, the students learn the theoretical background for their work in this module in the module IN8005 (lecture and exercise course and voluntary tutor-exercises).
Students work in groups of five on a practical programming problem (typically from the field of management) using a small database. According to the software development process, regular group meetings and meetings with the teaching staff take place in which the progress is monitored and assistance is given.

Media formats:
slides, problem specification sheets, moderated discussion boards in suitable e-learning platforms, (software development environment), (group and tutor meetings).

Literature:
• Learning Materials for IN8005 (continuously updated).
• S. Reges, M.Stepp: Building Java Programs: A Back to Basics Approach, Pearson 2014
• K. Rubin: Essential Scrum, Addison Wesley, 2012

Responsible for the module:
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Courses (Type, SH) Lecturer:

0000001833 Introduction to Informatics for Students of Management and Technology - Programming Lab (IN8027) (2SWS PR, WS 2024/25)
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For further information about this module and its allocation to the curriculum see:
https://campus.tum.de/tumonline/wbModHb.wbShowMHBReadOnly?pKnotenNr=1605334

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